Training a small network

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One complete restricted-training and evaluation record.

Trace repeated gradients/updates,repair objective scaling and explain the frozen representation limit.

Keep one habit: W1–W3/W5 → selected new training pair → project configuration ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.
- Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Use one hidden ReLU unit h = ReLU(x) and an **output** y = vh. The hidden **input** weight is fixed at 1, its bias is fixed at 0, and the **output** bias is fixed at 0. Only **output** weight v is trained.

  This is a restricted two-layer network, not **training** every parameter of a general network.

- **Training** pairs are (x, target) = (1, 2) and (2, 4). Both x values are positive, so hidden activations are 1 and 2. Define ordinary MSE over both rows.

  Start at v = 0 and use learning rate 0.1 for exactly four full-data updates. The declared budget prevents stopping only when a preferred result appears.

- Each row evaluates the parameter value after that many updates. Row zero records the starting model. The loss decreases in this specific experiment; that pattern is not a promise for all networks or rates.

Updating parameters

> **My remembered idea, example and correction**

Slopes and local change

My remembered idea, example and correction

Forward passes by hand

My remembered idea, example and correction

Lesson reference for workbook practice

Update count	v	Training MSE
0	0	10
1	1	2.5
2	1.5	0.625
3	1.75	0.15625
4	1.875	0.0390625

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main hReLU(x), outputvh; hiddenweight 1/hiddenbias 0/outputbias 0 frozen, onlyvtrained. Training(1, 2)/(2, 4), MSEaverage2 rows, startv 0, rate.1, exactly 4 full-dataupdates. States v0/1/1.5/1.75/1.875; MSE 10/2.5/.625/.15625/.0390625. Gradients5(v−2):−10/−5/−2.5/−1.25 forfourupdates. Atv1, SSE 5 butmean-gradient−5, not sum-gradient−10. Frozenv1.875 reserved(3, 6)prediction 5.625, error.140625.

  If proposed tasky 2 xincludes−1, thefixedh 0 predicts0 anderror4 foranyv. Separatepictureonepairx3, t3, rate 1/18 fitsv1; doublingtargetfitsv 2. Exact training fit alone proves neither memorization nor generalization. Independentpairuses its own rate/budget.

- At v = 0, calculate both **prediction**s, errors, MSE and gradient, then update at rate 0.1.

> **My reasoning and evidence**
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> |---|
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MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main hReLU(x), outputvh; hiddenweight 1/hiddenbias 0/outputbias 0 frozen, onlyvtrained. Training(1, 2)/(2, 4), MSEaverage2 rows, startv 0, rate.1, exactly 4 full-dataupdates. States v0/1/1.5/1.75/1.875; MSE 10/2.5/.625/.15625/.0390625. Gradients5(v−2):−10/−5/−2.5/−1.25 forfourupdates. Atv1, SSE 5 butmean-gradient−5, not sum-gradient−10. Frozenv1.875 reserved(3, 6)prediction 5.625, error.140625.

  If proposed tasky 2 xincludes−1, thefixedh 0 predicts0 anderror4 foranyv. Separatepictureonepairx3, t3, rate 1/18 fitsv1; doublingtargetfitsv 2. Exact training fit alone proves neither memorization nor generalization. Independentpairuses its own rate/budget.

- At v = 1, the sum of squared errors is 5. A learner uses gradient -10 with rate 0.1. Explain the mismatch with this lesson and give the next v under MSE.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main hReLU(x), outputvh; hiddenweight 1/hiddenbias 0/outputbias 0 frozen, onlyvtrained. Training(1, 2)/(2, 4), MSEaverage2 rows, startv 0, rate.1, exactly 4 full-dataupdates. States v0/1/1.5/1.75/1.875; MSE 10/2.5/.625/.15625/.0390625. Gradients5(v−2):−10/−5/−2.5/−1.25 forfourupdates. Atv1, SSE 5 butmean-gradient−5, not sum-gradient−10. Frozenv1.875 reserved(3, 6)prediction 5.625, error.140625.

  If proposed tasky 2 xincludes−1, thefixedh 0 predicts0 anderror4 foranyv. Separatepictureonepairx3, t3, rate 1/18 fitsv1; doublingtargetfitsv 2. Exact training fit alone proves neither memorization nor generalization. Independentpairuses its own rate/budget.

- At final v = 1.875, calculate the **prediction** and squared error on reserved pair (3, 6). May its target be included in the four reported **training** updates?

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main hReLU(x), outputvh; hiddenweight 1/hiddenbias 0/outputbias 0 frozen, onlyvtrained. Training(1, 2)/(2, 4), MSEaverage2 rows, startv 0, rate.1, exactly 4 full-dataupdates. States v0/1/1.5/1.75/1.875; MSE 10/2.5/.625/.15625/.0390625. Gradients5(v−2):−10/−5/−2.5/−1.25 forfourupdates. Atv1, SSE 5 butmean-gradient−5, not sum-gradient−10. Frozenv1.875 reserved(3, 6)prediction 5.625, error.140625.

  If proposed tasky 2 xincludes−1, thefixedh 0 predicts0 anderror4 foranyv. Separatepictureonepairx3, t3, rate 1/18 fitsv1; doublingtargetfitsv 2. Exact training fit alone proves neither memorization nor generalization. Independentpairuses its own rate/budget.

- For the lesson architecture at x = -1, give h and **prediction** for any finite v. If target is -2, calculate squared error and explain why changing v cannot fix it.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Keep h = ReLU(x), train only v, and use one pair (2, 2), v = 0 and rate 0.125. Use squared error and gradient 2 times (2v - 2) times 2. Perform two updates, reporting each new v and loss.

  Explain why the second update changes nothing.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Run a new training case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled **training** a small network **evidence** sheet to Image-pattern Network Investigation. Show how you can trace repeated forward loss gradient and update steps.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Check the mean convention for **Training** a small network. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Training** a small network. Proposed explanation: "Every parameter was trained". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Keep h = ReLU(x), train only v, and use one pair (2, 2), v = 0 and rate 0.125. Use squared error and gradient 2 times (2v - 2) times 2. Perform two updates, reporting each new v and loss.

  Explain why the second update changes nothing.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Train a single **output** weight while clearly marking the frozen hidden computation.
- Toy h=ReLU(x), y=vh. One **training** pair x = 3, target 3; start v = 0; squared loss; rate 1/18.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/12

Training a small network

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Freeze hidden step

h=3 for this pair; only v is trainable.



2. Compute initial gradient

Loss 9; gradient 2(3v-3) x 3=-18 at v=0.



3. Update and verify

v becomes 1; prediction 3 and loss 0. Next gradient is 0.

CHANGE ONE THING

Change target from 3 to6 and restart v = 0 with the same rate.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Train a single **output** weight while clearly marking the frozen hidden computation.
- Which labelled step matches this detail? v becomes 1; **prediction** 3 and loss 0. Next gradient is 0.
- 1. Freeze hidden step
- 2. Update and verify

- 3. Compute initial gradient

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Update and verify v becomes 1; **prediction** 3 and loss 0. Next gradient is 0. Only the **output** weight v is being fitted in this toy network; the hidden ReLU is fixed.

The **training input** gives hidden value three. Squared-error slope points v from zero towards one, and this specially chosen rate reaches one in a single update.

Zero loss on one **training** pair is exact **fitting** of that pair, not **evidence** about other **input**s.

Changed-case question and optional hint

- Changed-case practice: Change target from 3 to6 and restart v = 0 with the same rate. Which conclusion is supported by the supplied facts?

- 1. First gradient=-36; v becomes 2; **prediction** 6 and loss 0.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: First gradient=-36; v becomes 2; **prediction** 6 and loss 0. The hidden value remains 3, but the target residual and required **output** weight double.

Repair and retain the record

Review Updating parameters (ai-5/11). Goal: Apply a learning rate to an ordered gradient. Short practice: Why must new loss be recomputed rather than assumed smaller?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

Choose the picture that helps explain the current step.

01. Training a small network: the case

AI ACADEMY / AI-5/12 · 01 · LOOK AT THE FACTS

Training a small network: the case

The situation

Train a single output weight while clearly marking the frozen hidden computation.

Supplied evidence

Toy h=ReLU(x), y=vh. One training pair x = 3, target 3; start v = 0; squared loss; rate 1/18.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain training a small network using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

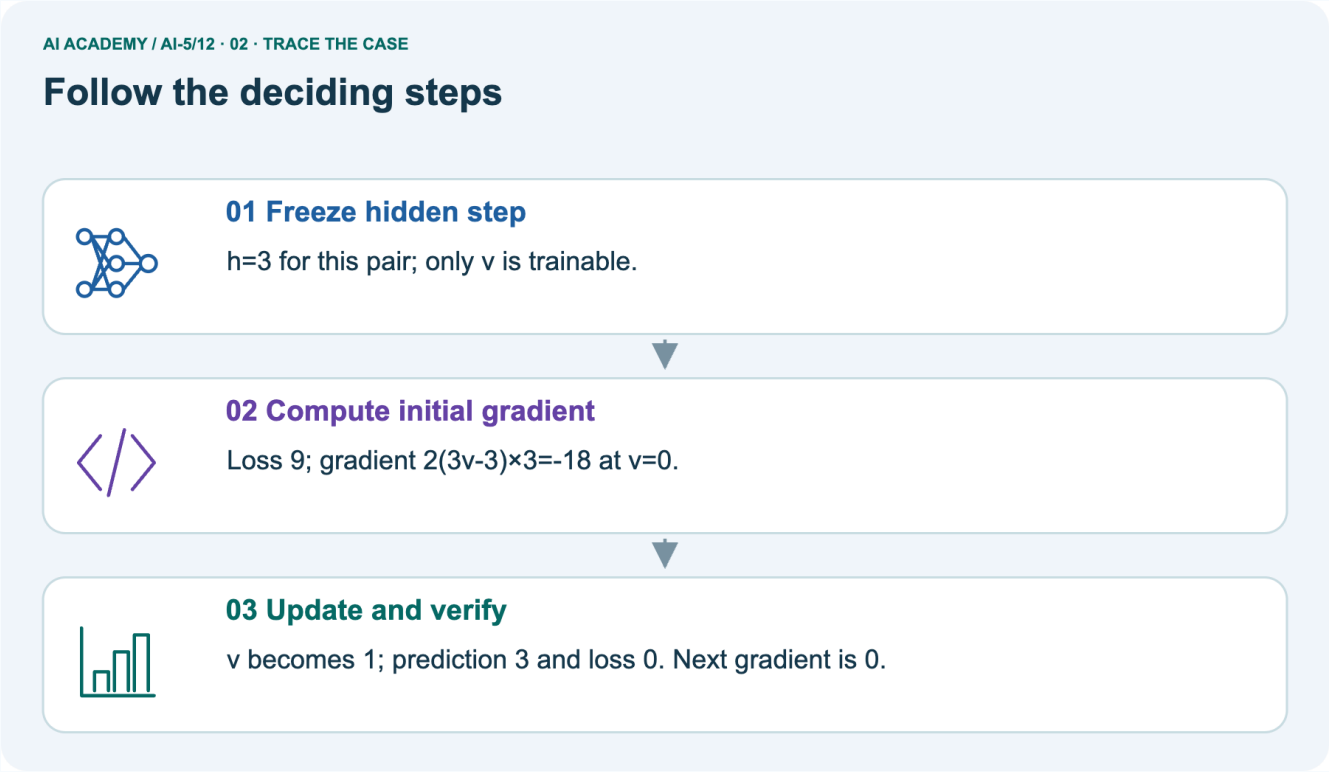
Train a single **output** weight while clearly marking the frozen hidden computation. Supplied facts: Toy h=ReLU(x), y=vh. One **training** pair x = 3, target 3; start v = 0; squared loss; rate 1/18.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze hidden step: h=3 for this pair; only v is trainable. → Compute initial gradient: Loss 9; gradient 2(3v-3)×3=-18 at v=0. → Update and verify: v becomes 1; **prediction** 3 and loss 0. Next gradient is 0.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. One full-data update from v is 1

AI ACADEMY / AI-5/12 · 03 · INSPECT THE LESSON MODEL

One full-data update from v is 1

Stage	Compute with current v	Result
Forward and loss	Predictions [1, 2] for targets [2, 4]	Errors [-1, -2], MSE 2.5
Gradient	2 times (-1 times 1 - 2 times 2) / 2	Gradient -5
Update with rate 0.1	1 - 0.1 times -5	Next v is 1.5

Each row depends on the preceding calculation. Recompute predictions and loss on the next iteration; do not keep using the first gradient after parameters change.

Follow the labelled picture with the matching explanation. Open the vector diagram

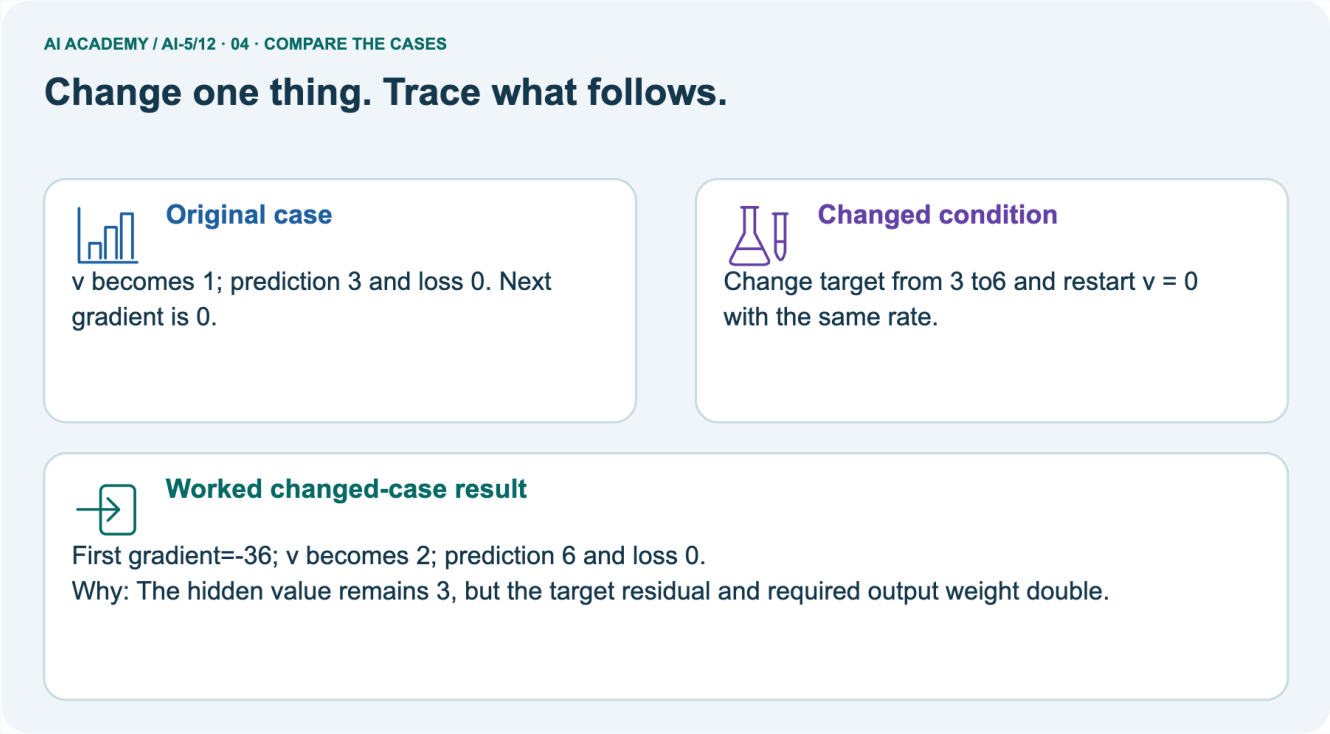
One full-data update from v is 1. Forward and loss | **Prediction**s [1, 2] for targets [2, 4] | Errors [-1, -2], MSE 2.5. Gradient | 2 times (-1 times 1 - 2 times 2) / 2 | Gradient -5.

Update with rate 0.1 | 1 - 0.1 times -5 | Next v is 1.5

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

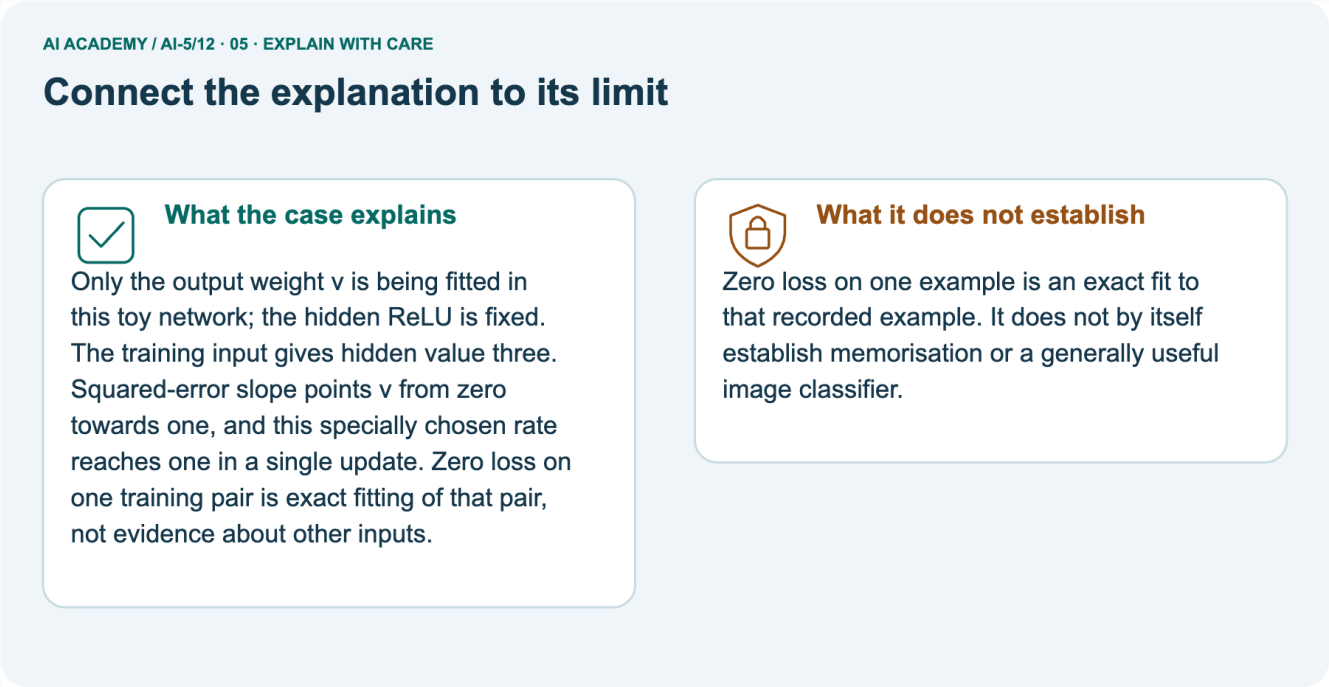
Original case: v becomes 1; **prediction** 3 and loss 0. Next gradient is 0. Changed condition: Change target from 3 to6 and restart v = 0 with the same rate.

Expected result: First gradient=-36; v becomes 2; **prediction** 6 and loss 0. Reason: The hidden value remains 3, but the target residual and required **output** weight double.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



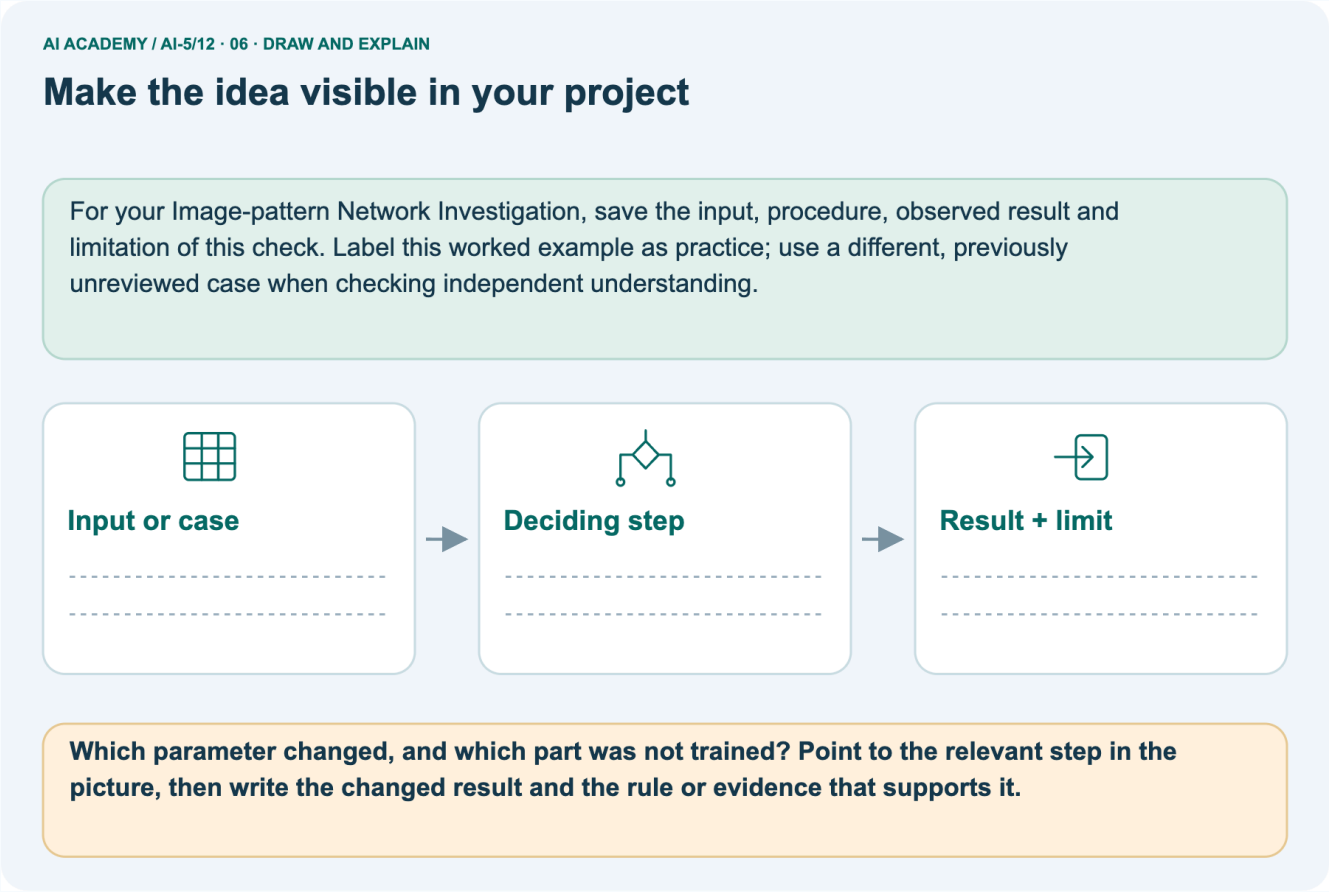
Follow the labelled picture with the matching explanation. Open the vector diagram

Only the **output** weight v is being fitted in this toy network; the hidden ReLU is fixed. The **training input** gives hidden value three. Squared-error slope points v from zero towards one, and this specially chosen rate reaches one in a single update. Zero loss on one **training** pair is exact **fitting** of that pair, not **evidence** about other **input**s. Limit: Zero loss on one example is an exact fit to that recorded example. It does not by itself establish memorisation or a generally useful image classifier.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Training a small network: the case

02. Follow the deciding steps

03. One full-data update from v is 1

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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